



University of San Carlos | Department of
COMPUTER ENGINEERING

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P2

CpE 3104 – Microprocessors

Memory Address Decoding and Interfacing Part II

Unit 5: MEMORY INTERFACING

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Outline

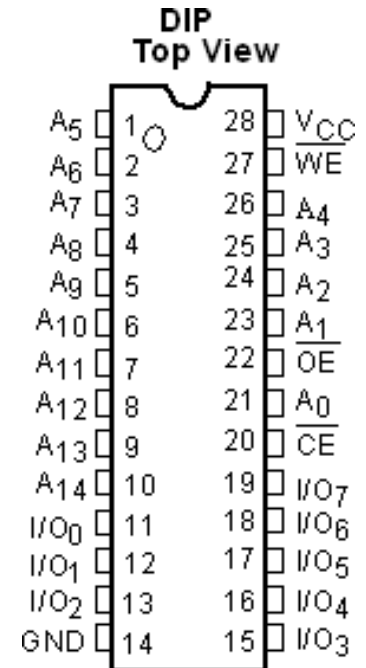


P2

- Interfacing 8-bit Memory Chips to 8086 μ P
- Bank Selector
- Memory Access Control Signals
- Address Decoding for Memory Banks (RAM)
- Address Decoding for Memory Banks (ROM)
- Bank Selection
- Why Not Connect A_0 to Memory Chips?
- Why Latch the Memory Address?

Interfacing 8-bit Memory Chips to 8086 μ P

- Most memory chips designed for the 8086 microprocessor has 8-bit data bus.
- The available memory chips are 8-bit like the 62256 SRAM and 2764 EPROM.

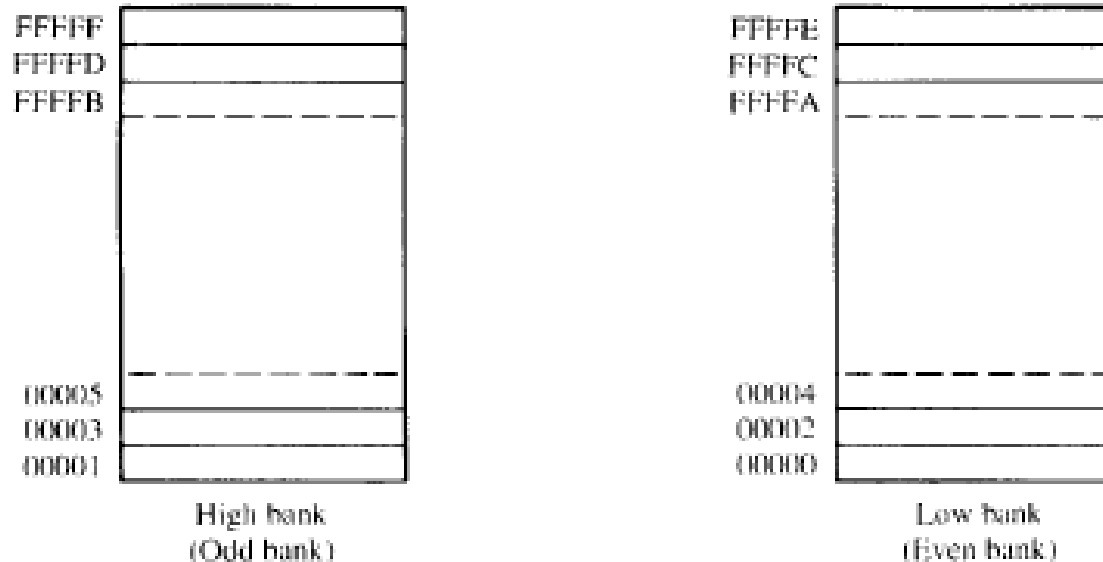


62256 - 32K x 8 Static RAM

Interfacing 8-bit Memory Chips to 8086 μ P

- The data bus of the 8086 microprocessor presents a problem since the microprocessor has a 16-bit data bus. The 8086 μ P need to write 16-bit or 8-bit data.
- This means that the 16-bit data bus must be divided into two separate sections (or banks) that are 8 bits wide so that the microprocessor can write to either half (8 bits) or both halves (16 bits).

Interfacing 8-bit Memory Chips to 8086 μ P



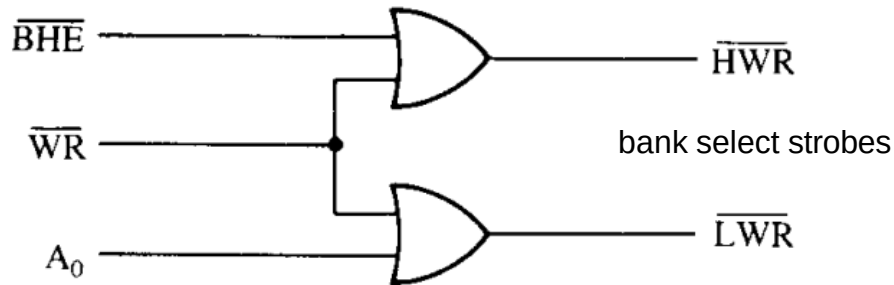
Note:

The **upper byte** of a 16-bit data will be stored to **High bank** at an “**odd**” address while the **lower byte** will be stored to the **Low bank** at an “**even**” address.

Bank Selector

- For the 8086 μ P, the bank selector is **BHE** and it goes **low** when it **writes** to the **high bank**, otherwise, it **writes** to the **lower bank**.
- However, the bank selection is important only when writing data to RAM. Therefore, the WR signal should be the primary signal since bank selection happens only during write operation.
- A separate **bank select signal strobes** will be created.

Bank Selector



$\overline{BHE}$	$\overline{BLE}$	Function
0	0	Both banks enabled for a 16-bit transfer
0	1	High bank enabled for an 8-bit transfer
1	0	Low bank enabled for an 8-bit transfer
1	1	No bank enabled

A_0 is the address check whether the address is odd or even. If $A_0 = 0$ then the data is to be written to the **Low bank (even)**, thus, **LWR will be logic-0**, therefore, **enabling** only the **Low bank** RAM chips.

LWR and HWR will not be logic-0 at the same time.

Note: A_0 will not be connected to data lines of the RAM chip.
 (Click [here](#) for more information.)

Memory Access Control Signals

$\overline{\text{RD}}$

During a memory read operation, the signal will be logic-0. Use this signal to enable the output of both the RAM and the ROM chips.

$\overline{\text{WR}}$

During a memory write operation, the signal will be logic-0. Use this signal to select banks with A_0 and $\overline{\text{BHE}}$ producing the select strobes $\overline{\text{LWR}}$ and $\overline{\text{HWR}}$. The select strobes shall be connected to the write enable pins of the high bank or low bank RAM chips. This is not available for the ROM chips.

$\text{M}/\overline{\text{IO}}$

During a memory read or write operation, the signal will be logic-1. If accessing isolated I/O using instructions IN or OUT, the signal will be logic-0. This is also used to decode I/O address.

Address Decoding for Memory Banks (RAM)

- The number of chips must be determined for both banks.
- From the number of chips, the number of sets can be determined. Each set contains an even (low) and odd (high) banks.
- Address ranges for each set can be calculated. It will be used to design the decoder circuit.

Address Decoding for Memory Banks (RAM)

For example, if the RAM size is 64K x 16, it can be constructed using 32K x 8 chips.

$$\text{no. of chips} = \frac{\text{RAM Size}}{\text{chip size}} \times 2 \text{ banks} = \frac{64\text{K}}{32\text{K}} \times 2 \text{ banks} = 4 \text{ chips}$$

$$\text{no. of sets}^* = \frac{\text{no. of chips}}{2 \text{ banks}} = \frac{4}{2} = 2 \text{ sets}$$

Note:

The **address range** of *each set* will be determined instead of the *individual chips*.

* A set is composed of a Low and High bank chips

Address Decoding for Memory Banks (RAM)

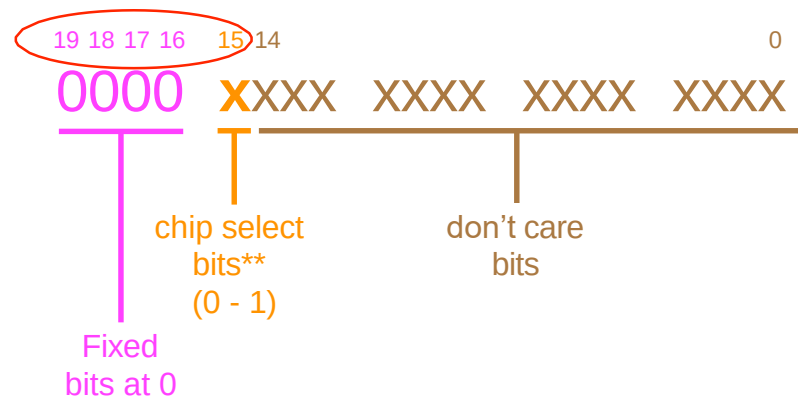
start address* = end address of previous set + 1

end address = (start address of set) + (chip size – 1)

(chip size – 1): 32K – 1 = 07FFFH

Set	Start Address	End Address	Address Range
1	00000H	00000H + 07FFFH = 07FFFH	00000H - 07FFFH
2	07FFFH + 1 = 08000H	08000H + 07FFFH = 0FFFFH	08000H - 0FFFFH

Set 1	0000	00	00	0000	0000	0000
	0000	01	11	1111	1111	1111
Set 2	0000	10	00	0000	0000	0000
	0000	11	11	1111	1111	1111



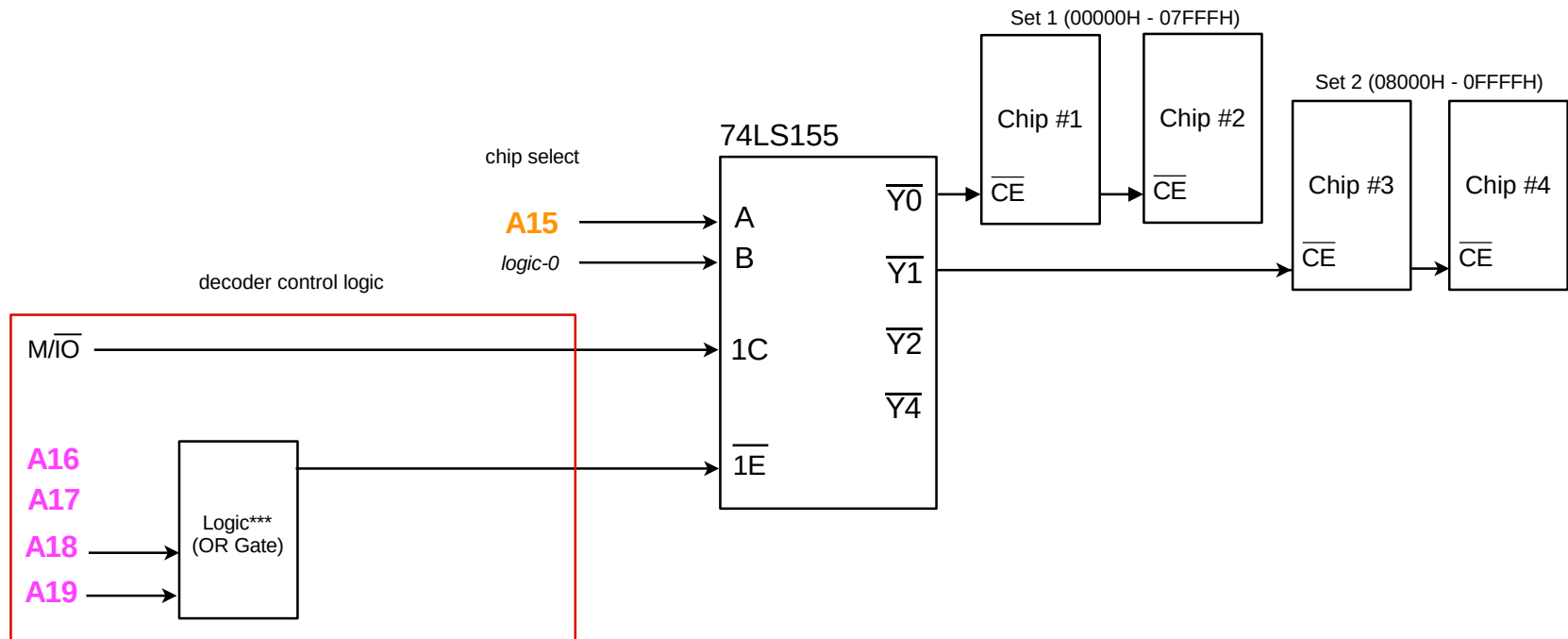
* Start address of the first set is 00000H (see 8086 memory map).

Address Decoding for Memory Banks (RAM)

Since there are **2** (2^1) **sets**, we need to decode **1 bit to 2**.

A $2 \times 4^*$ decoder will be used such as the 74LS155.

The **chip select** bits will be used as the input and the **fixed** bits as the control for the 74LS155.



* No available 1×2 decoder IC.

** M/IO pin will be logic-0 when accessing memory.

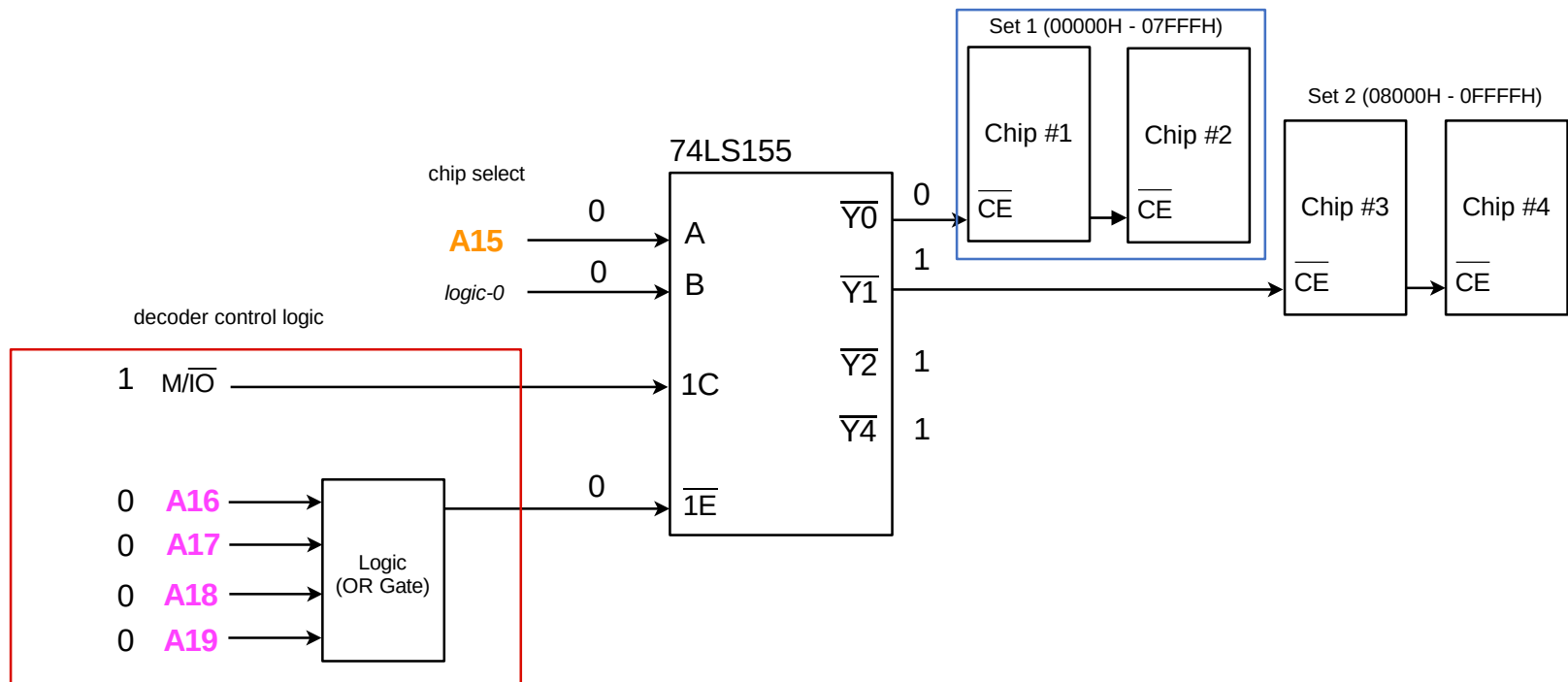
*** Logic function depends on the control signal requirement of the $n \times 2^n$ decoder IC.

Address Decoding for Memory Banks (RAM)

For example, if the memory address is **03F00H**:

0000 0011 1111 0000 0000

Address is **valid** since it is between the RAM address range.

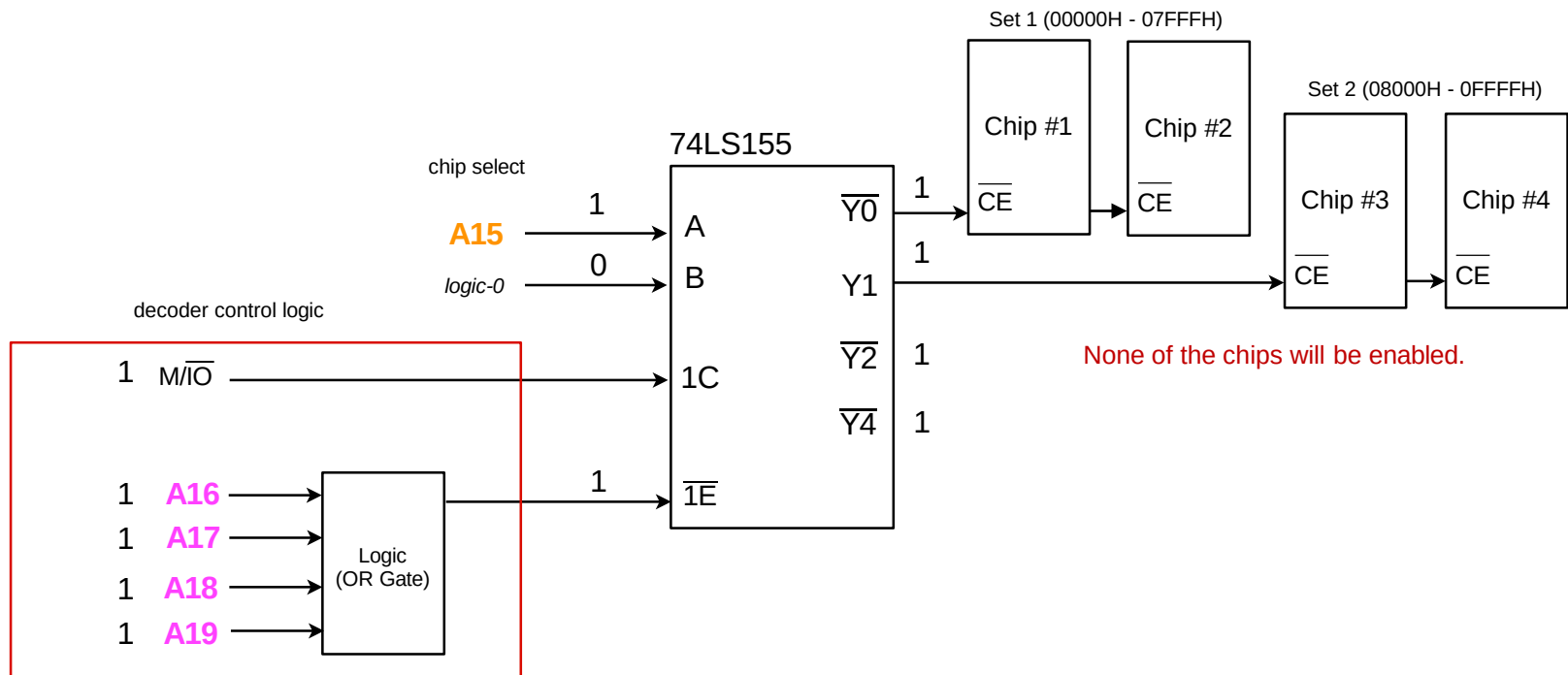


Address Decoding for Memory Banks (RAM)

For example, if the memory address is **FF000H**:

1111 **1**111 0000 0000 0000

Address is **invalid** since it is outside the RAM address range.



Address Decoding for Memory Banks (ROM)

- The process is similar to address decoding with RAM especially if both have the **same size and chip count**.
- Since the ROM is on the upper portion of the memory map, the **fixed bits** are usually logic-1.

Address Decoding for Memory Banks (ROM)

For example, if the ROM size is 32K x 16, it can be constructed using 8K x 8 chips.

$$\text{no. of chips} = \frac{\text{ROM Size}}{\text{chip size}} \times 2 \text{ banks} = \frac{32\text{K}}{8\text{K}} \times 2 \text{ banks} = \mathbf{8 \text{ chips}}$$

$$\text{no. of sets}^* = \frac{\text{no. of chips}}{2 \text{ banks}} = \frac{8}{2} = \mathbf{4 \text{ sets}}$$

Note:

The address range of each set will be determined instead of the individual chips.

* A set is composed of a Low and High bank chips.

Address Decoding for Memory Banks (ROM)

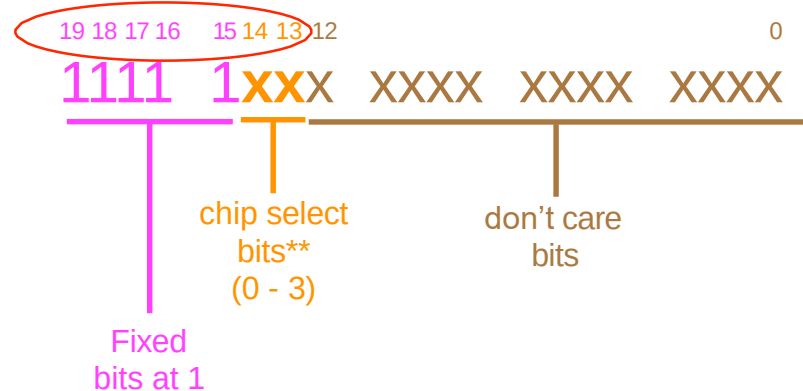
end address = (start address of previous set -1)

start address* = end address of set – (chip size – 1)

(chip size – 1): 8K - 1 = 01FFFFH

Set	Start Address	End Address	Address Range
1	FFFFFFH - 01FFFFH = FE000H	FFFFFFH	FE000H - FFFFFFH
2	FDFFFFH - 01FFFFH = FC000H	FE000H - 1 = FDFFFFH	FC000H - FDFFFFH
3	FBFFFFH - 01FFFFH = FA000H	FC000H - 1 = FBFFFFH	FA000H - FBFFFFH
4	F9FFFFH - 01FFFFH = F8000H	FA000H - 1 = F9FFFFH	F8000H - F9FFFFH

Set 4	1111	1000	0000	0000	0000
	1111	1001	1111	1111	1111
Set 3	1111	1010	0000	0000	0000
	1111	1011	1111	1111	1111
Set 2	1111	1100	0000	0000	0000
	1111	1101	1111	1111	1111
Set 1	1111	1110	0000	0000	0000
	1111	1111	1111	1111	1111



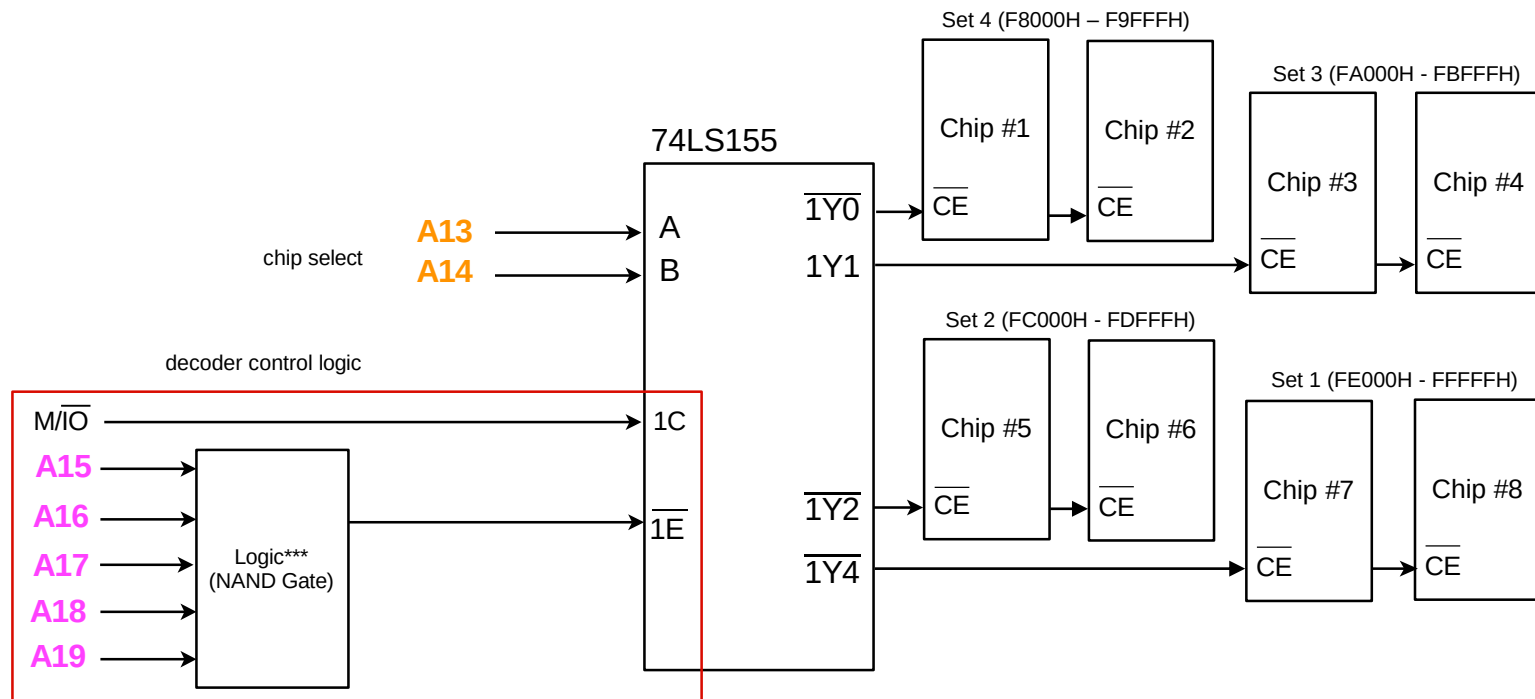
* End address of the first set is FFFFFH (see 8086 memory map).

Address Decoding for Memory Banks (ROM)

Since there are 4 (2^2) sets, we need to **decode 2 bits to 4**.

A 2 x 4 decoder will be used such as the 74LS155.

The **chip select** bits will be used as the input and the **fixed** bits as the control for the 74LS155*.



* It has an extra control pin compared to 74LS139 which adds control flexibility.

** M/I $\overline{O}$ pin will be logic-0 when accessing memory.

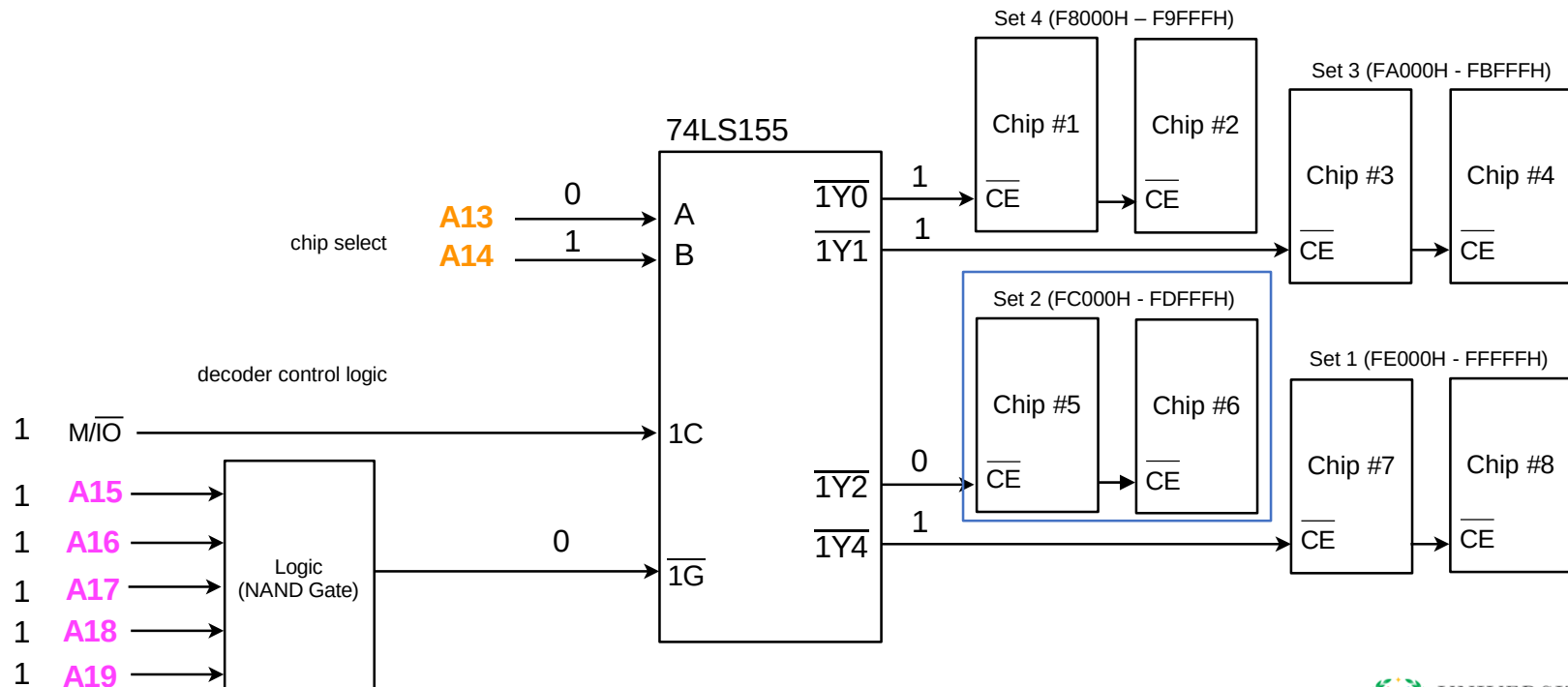
*** Logic function depends on the control signal requirement of the $n \times 2^n$ decoder IC.

Address Decoding for Memory Banks (ROM)

For example, if the memory address is **FC005H**:

1111 1100 0000 0000 0101

Address is **valid** since it is between the ROM address range.

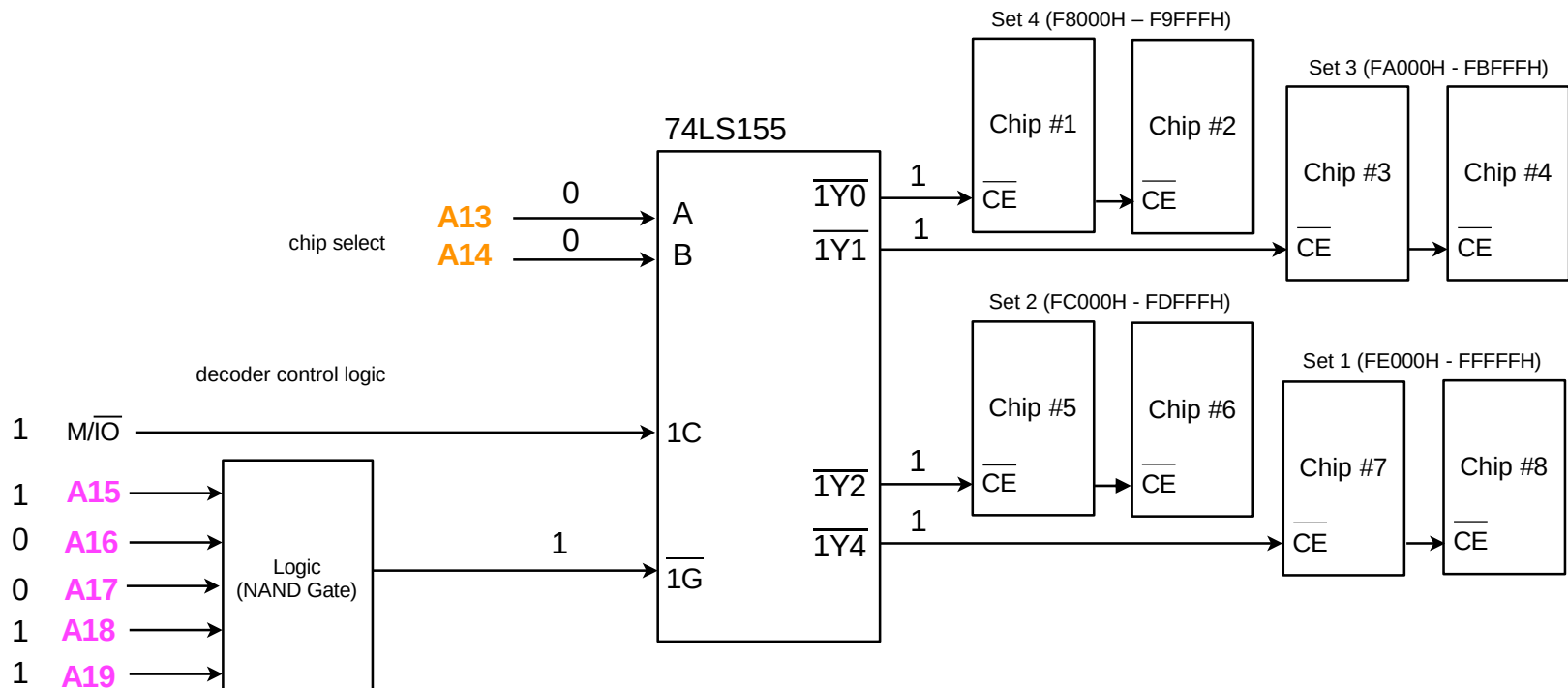


Address Decoding for Memory Banks (ROM)

For example, if the memory address is **C8207H**:

1100 1000 0010 0000 0111

Address is **invalid** since it is outside the ROM address range.



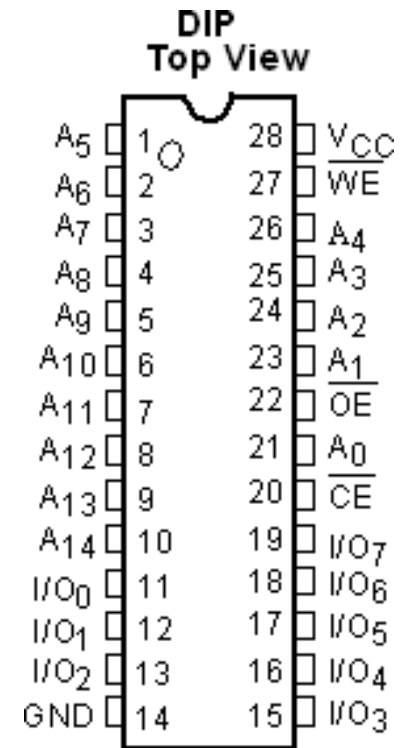
Since the decoder is disabled, none of the chips should be enabled.

Bank Selection

- Bank selection for write operation is controlled by LWR (low bank) and HWR (high bank).
- The above control signals will be connected to the write enable pins of the RAM chips. Does not apply to ROM (since it is read-only*).
- **Example:** 62256 (32K x 8) SRAM

Bank Selection

- It has **15-bit address bus** ($A_0 - A_{15}$) and **8-bit data bus** ($I/O_0 - I/O_7$)
- **Control pins** include:
 - $\overline{CE}$ or $\overline{CS}^*$
 - enables the chip, allows read or write operation
 - $\overline{OE}$
 - enables output on data bus for read operation
 - $\overline{WE}$
 - enables write operation
- $\overline{WE}$ will be used in the bank selection.
- A_0 of the 8086 address bus will not be connected to the chip.



62256 - 32K x 8 Static RAM

* Used in some literature or schematic symbol.

Bank Selection

Control and Bus Signals

From	Signal	Connect to	Purpose
Bank Select Circuit	$\overline{\text{HWR}}$	$\overline{\text{WE}}$ of all high bank RAM chips	Selects which bank to enable either low or high banks during write operation only
	$\overline{\text{LWR}}$	$\overline{\text{WE}}$ of all low bank RAM chips	
Address Decoding Circuit	$\overline{\text{Y}}_0, \overline{\text{Y}}_1 \dots \overline{\text{Y}}_n$	$\overline{\text{CS}}$ or $\overline{\text{CE}}$ of both banks	Enables or disables a set (both banks); overrides the other control signals
Address Bus Buffer Circuit	$\text{A}_1 - \text{A}_{19}$	address bus ($\text{A}_0 - \text{A}_n$) of all RAM and ROM chips	Memory address of the data
Microprocessor	$\overline{\text{RD}}$	$\overline{\text{OE}}$ of all RAM and ROM Chips	Enables the output of the RAM and ROM chips during read operation only
	$\overline{\text{MIO}}$	connect to address decoding circuit	Enables the address decoder for when accessing memory
	$\text{AD}_0 - \text{AD}_{15}$	$\text{AD}_0 - \text{AD}_7$ to low bank of RAM and ROM chips	8-bit data read/write
		$\text{AD}_8 - \text{AD}_{15}$ to high bank of RAM and ROM chips	16-bit data read-write in conjunction with $\text{AD}_0 - \text{AD}_7$

Why Not Connect A_0 to Memory Chips?

- A_0 is already connected to bank select circuit.
- If A_0 is connected to the address pin of memory chips, every other location in each bank of memory will be used.
- This means that half of the memory is wasted if A_0 of the address bus is connected to A_0 of memory.

A theoretical microprocessor has the following specification:

Address Bus: 4-bits

Address Range: 0H - FH

Data Bus: 16-bits

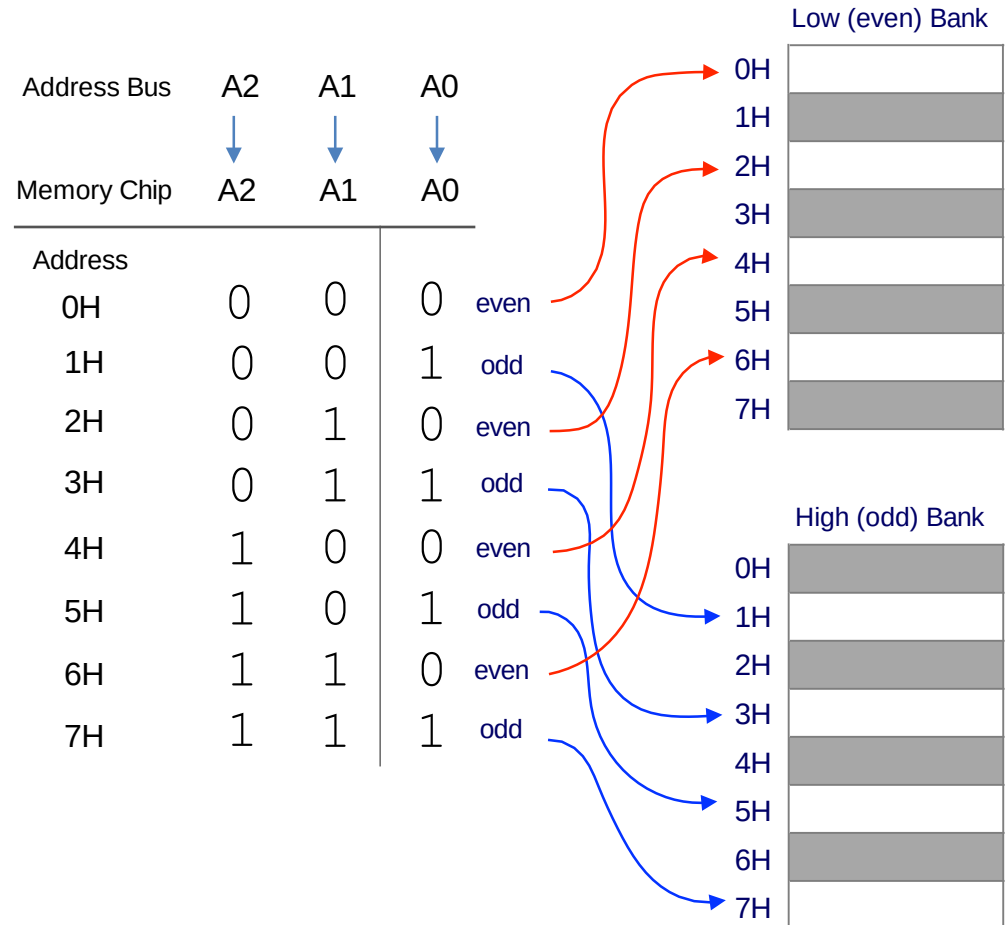
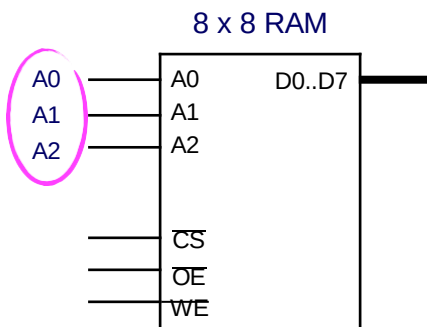
Using a theoretical RAM memory chip which has a size of 8 x 8bit.

Address Bus: 3-bits (A_0 - A_2)

Address Range: 0H - 7H

Data Bus: 8-bits

Since the microprocessor has a 16-bit data bus, the memory will be organized into **high** and **low** banks. In this example, A_0 of the address bus is connected to the chip.



Shaded areas in the memory are unreachable or unaddressable.

A theoretical microprocessor has the following specification:

Address Bus: 4-bits

Address Range: 0H - FH

Data Bus: 16-bits

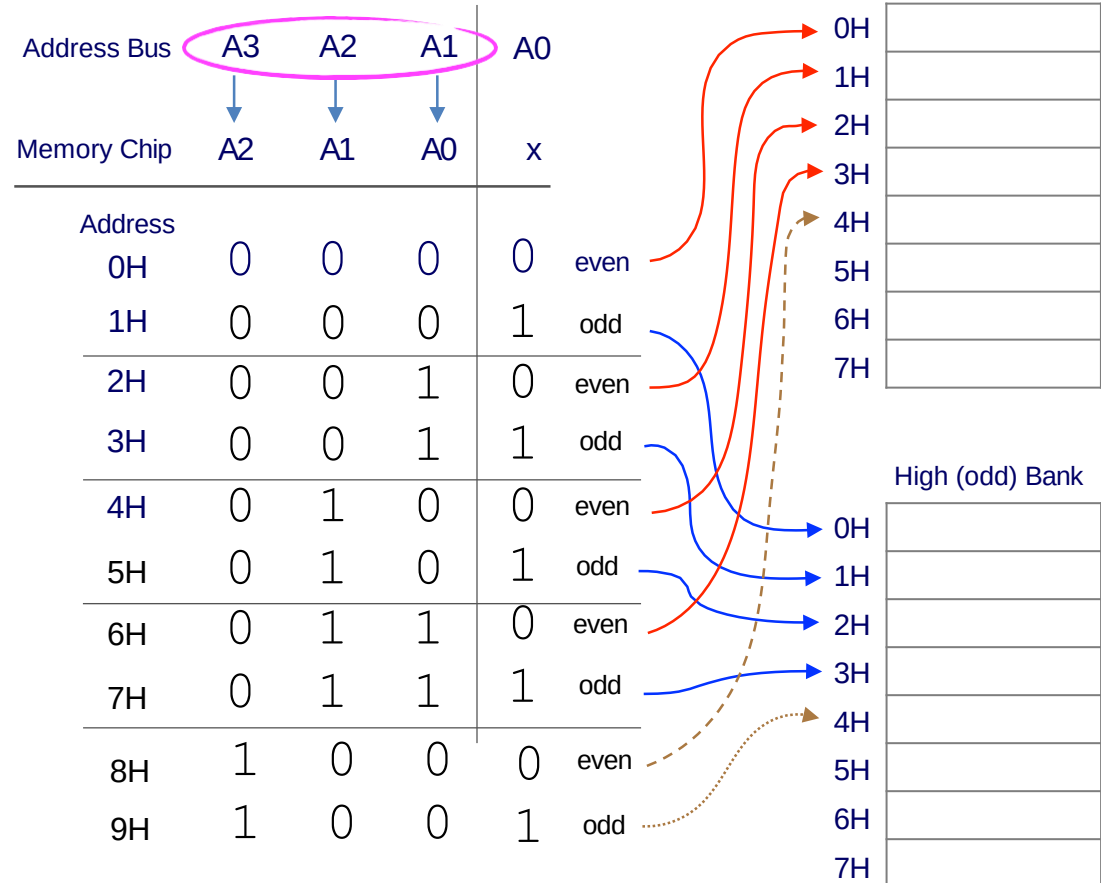
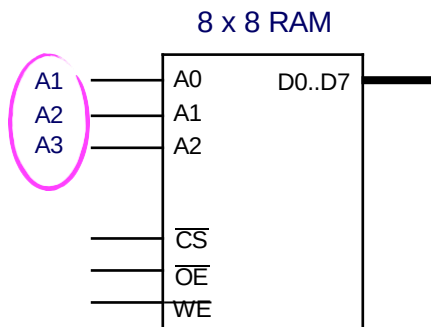
Using a theoretical RAM memory chip which has a size of 8 x 8bit.

Address Bus: 3-bits ($A_0 - A_2$)

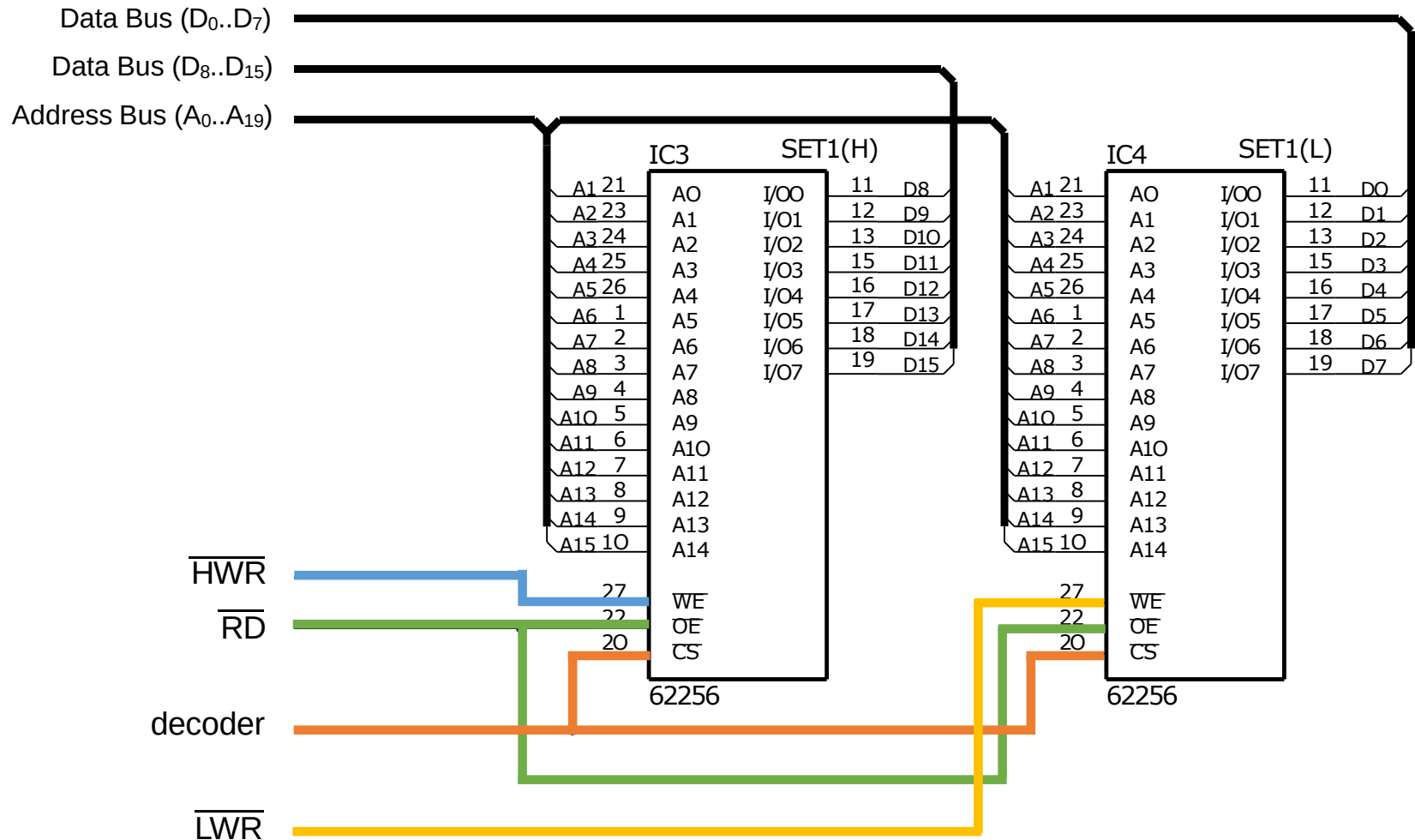
Address Range: 0H - 7H

Data Bus: 8-bits

Since the microprocessor has a 16-bit data bus, the memory will be organized into **high** and **low** banks with A_0 of the address bus not connected to the chip in this example.



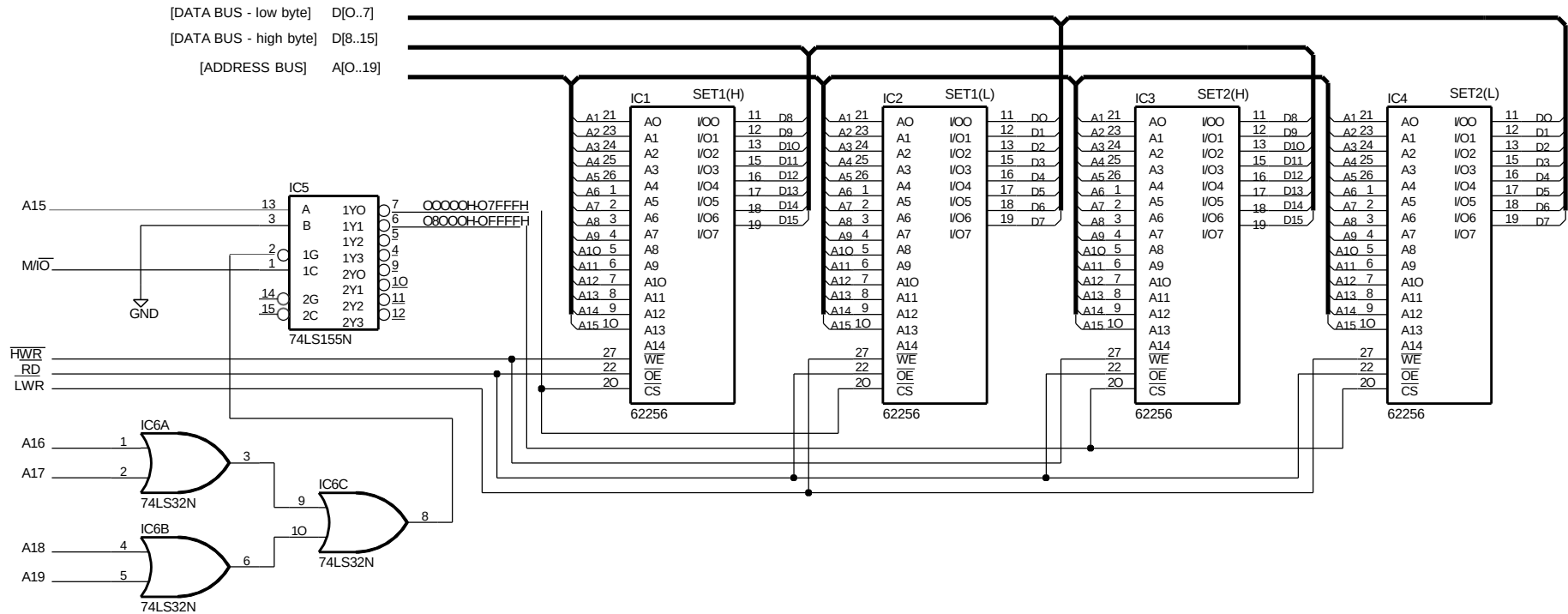
Address 4H - 7H can still be accessed by extending address to A_2 .



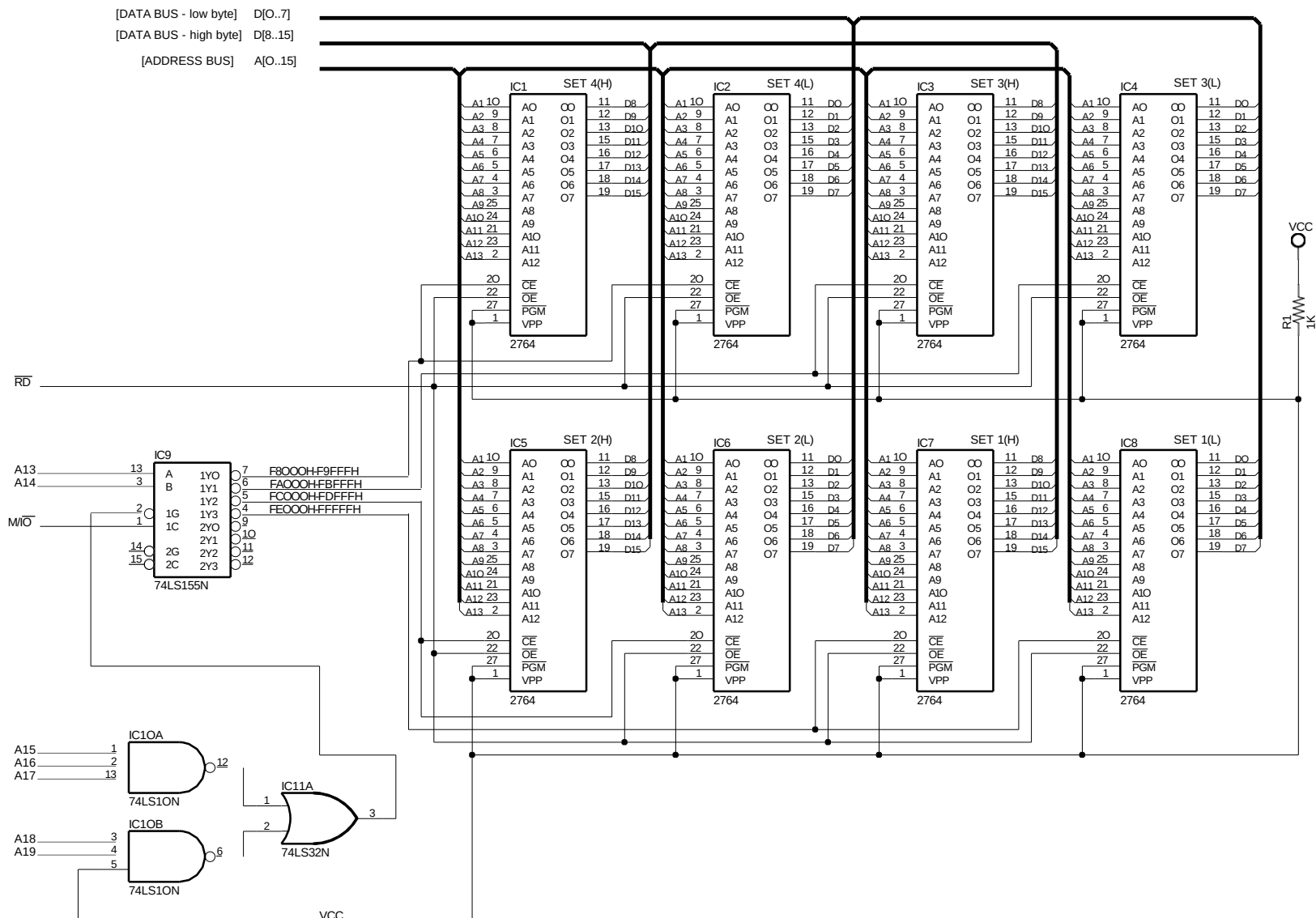
* A₀ of the address bus is not connected to the RAM and ROM chips. In this case pins A₀ - A₁₄ is connected to A₁ - A₁₅ of the address bus.

** HWR and LWR signals are applicable to ROM chips.

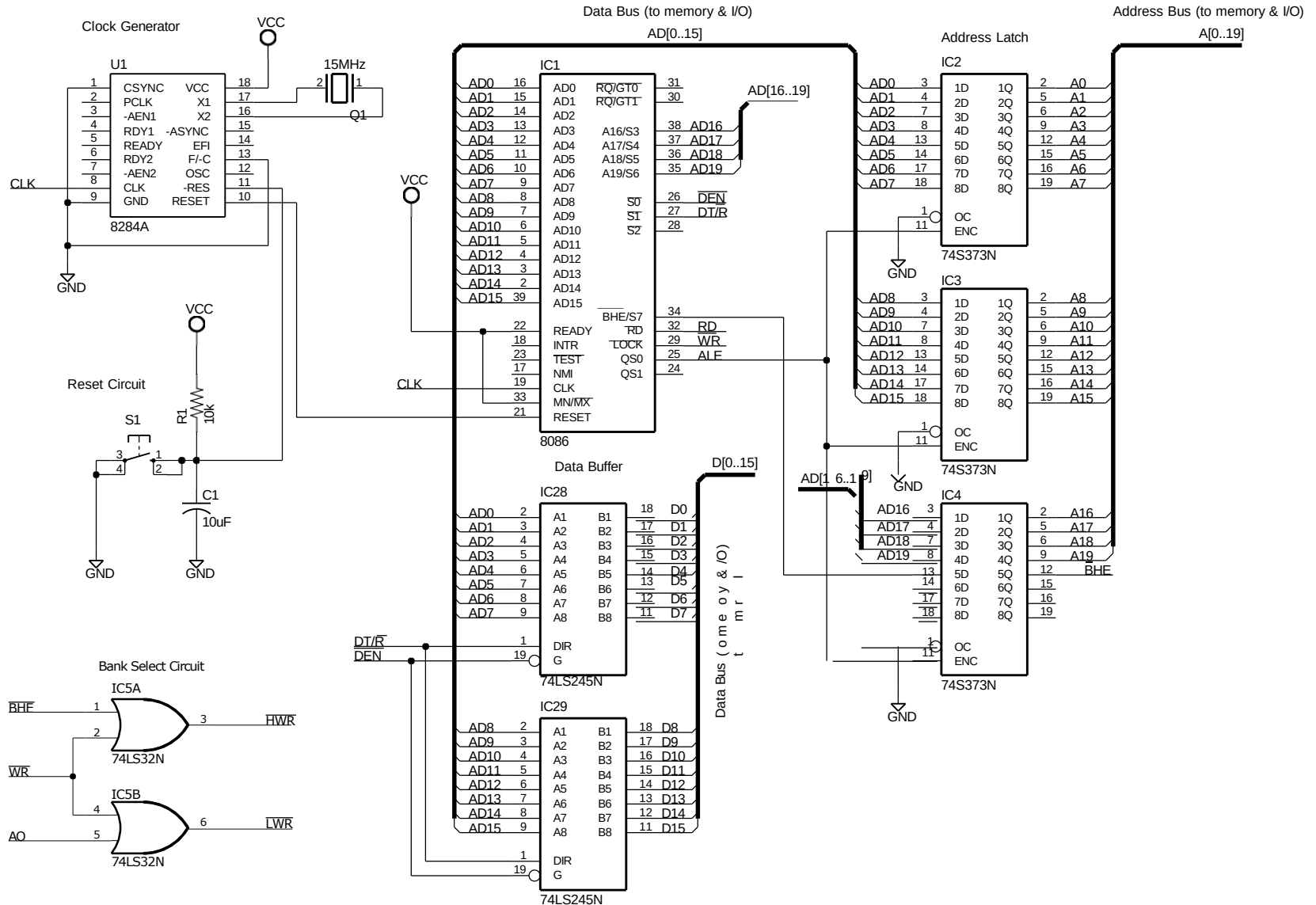
64K x 16 RAM using 4-62256 32K x 8 SRAM



32K x 16 ROM using 8-2764 8K x 8 EPROM



8086 with 8284 Clock Generator, Address Latch, Data Buffer and Bank Select Circuits

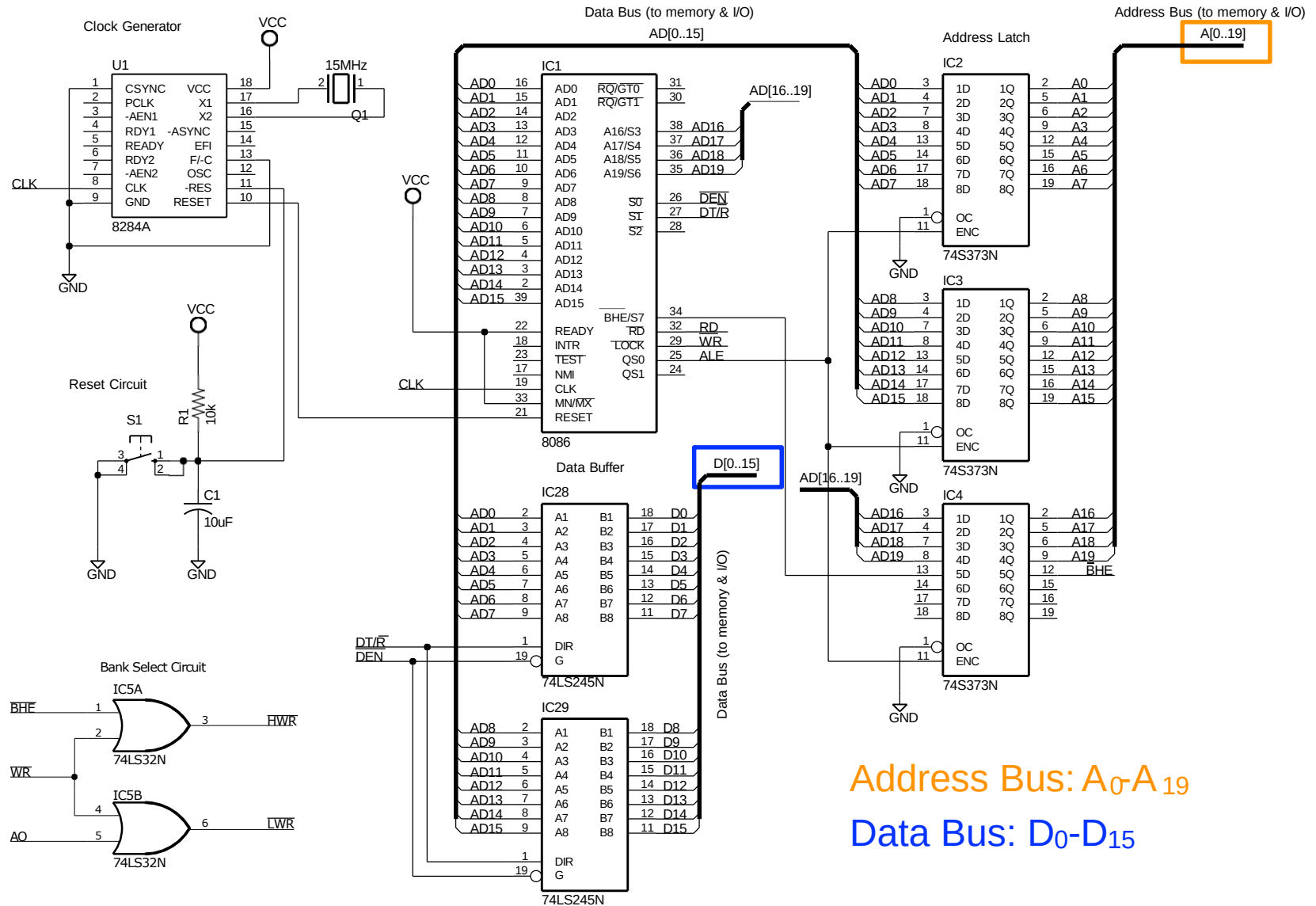


* The address/data bus of the 8086 is logically split in VSM, AD₀-AD₁₅ (16-bit data bus) and AD₁₆-AD₁₉.

Why Latch the Memory Address?

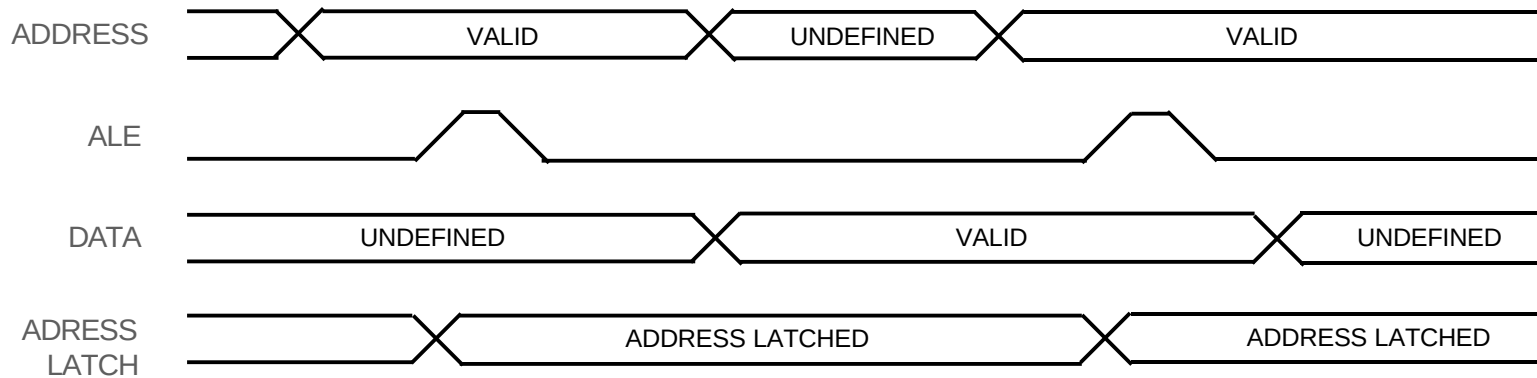
- The **address** and **data share** the **same physical bus**.
- When the microprocessor access memory or I/O, it has to write the address on the address bus.
- Once the microprocessor writes or reads data, the bus will now contain memory or I/O data, therefore, the address is now overwritten.
- Address on the memory chips will now change, thus, creating addressing problems.
- By **latching the memory address**, the **address is retained**.
- The addition of the **address and data buffer circuits** creates two separate channels for address and data.

8086 with 8284 Clock Generator, Address Latch, Data Buffer and Bank Select Circuits

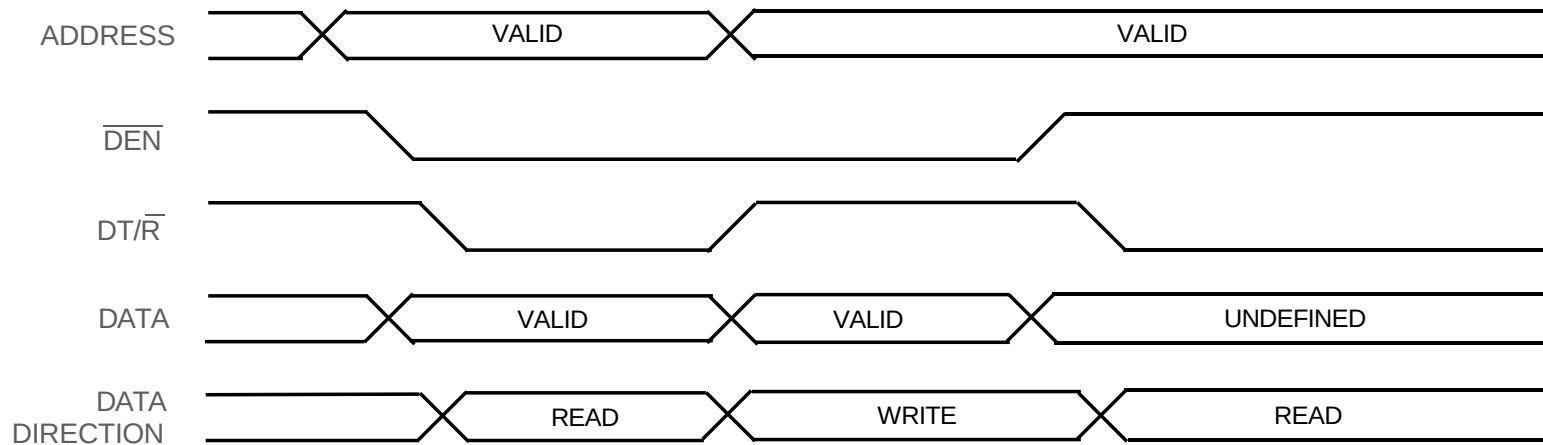


Address Bus: A₀-A₁₉

Data Bus: D₀-D₁₅



Address Latch Timing Diagram



Data Buffer/Direction Timing Diagram



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End of Lecture

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